

NEXUS-1

Operator & Engineering Interpretation Guide

How to read every screen, what the nuclear engineering means,
and how to respond to what it shows.

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Scope and honesty statement. NEXUS-1 is a single-file software demonstrator built to show software-engineering, simulation and visualisation skill applied to the nuclear-energy domain. It is **not** a certified operator-training simulator, confers **no qualification**, and has **no control authority** over any real plant. Some screens run rigorous physics; others are deliberately illustrative. This guide states which is which on every screen, because in this project honest framing is treated as a feature, not a caveat. The author is an Electrical & Software Engineer, not a nuclear engineer; the nuclear content here is the result of study, presented as interpretation, not authority.

How to use this guide

This guide is the *interpretation* layer of the NEXUS-1 documentation set. The Functional Guide tells you *what each page does* and where the controls are; the Glossary defines *terms*; the Reactor Physics Reference lists the *equations and constants*. This document stands in front of each screen with you and answers the question an operator or engineer actually asks: “*What am I looking at, what does it mean physically, what does good look like, and what do I do when it goes off-nominal?*”

The four-part template

Every full-depth screen entry is written in the same rhythm so you can scan it under pressure:

What you see

the panels, numbers and controls actually rendered on that screen, named as they appear.

The physics behind it

the reactor-engineering meaning of those numbers — why they exist and what governs them.

Normal vs. off-nominal

the signature of a healthy plant versus the early shapes of trouble, usually as a compact table.

Operator response

in a teal box — the reasoning an operator would apply. In NEXUS-1 most of this is observation and diagnosis, because the platform is advisory.

Legend of callouts

- **Teal box** — operator response / how to cope.
- **Amber box** — model-honesty note: this screen (or value) is illustrative, clamped, or representative rather than a validated plant figure.
- **Violet box** — a key concept worth carrying between screens.
- **Monospaced text** — a label, value or control exactly as it appears on screen.

The two physics codepaths

Model honesty – read this

NEXUS-1 contains *two* reactor models, and they do not use identical numbers. The **Reactor Kinetics** screen (and Training Mode, which reuses it) runs a rigorous six-group delayed-neutron point-kinetics solver. The **Neutronics** screen runs a lighter, illustrative reactivity-balance display whose feedback coefficients differ from the kinetics solver. This is a known, documented inconsistency; treat the Kinetics screen as the physically meaningful one and the Neutronics screen as a schematic. The roadmap’s next architectural step unifies both into one NEXUS.Physics module.

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1 Part I — Reactor fundamentals (the primer)

Read this once and the rest of the guide becomes self-explanatory. Everything on the reactor screens is an expression of the ideas below.

1.1 The chain reaction and k_{eff}

A reactor runs on a self-sustaining fission chain. Each fission of a ^{235}U nucleus releases energy and 2–3 fast neutrons; some of those go on to cause further fissions. The bookkeeping quantity is the *effective multiplication factor* k_{eff} : the ratio of neutrons in one generation to the previous one.

- $k_{\text{eff}} = 1$ — **critical**: population steady, power constant.
- $k_{\text{eff}} > 1$ — **supercritical**: power rising.
- $k_{\text{eff}} < 1$ — **subcritical**: power falling.

Criticality is *not* a synonym for danger — a reactor at steady full power is exactly critical. “Critical” simply means the chain is self-sustaining.

1.2 Reactivity: pcm and dollars

It is more convenient to talk about how far the core is *from* critical. That is reactivity, $\rho = (k_{\text{eff}} - 1)/k_{\text{eff}}$. Two units appear all over NEXUS-1:

- **pcm** (*per cent mille*, 10^{-5}): a fine-grained absolute unit. +10 pcm is a small nudge up; −1180 is the strong negative reactivity held by dissolved boron.
- **dollars (\$)**: reactivity normalised to the delayed-neutron fraction β . **\$1 of reactivity** = β . In the NEXUS-1 solver $\beta \approx 0.006502$, so **\$1 $\approx$ 650 pcm**. The Kinetics screen shows both, e.g. +325 pcm = \$0.50.

Why dollars matter

The dollar is the natural unit of *danger*, not just of size. Below \$1 the reactor is controllable on the slow timescale of delayed neutrons. **At \$1 the prompt neutrons alone sustain the chain (“prompt critical”), and power runs away on a millisecond timescale.** \$1 is the line you never cross. NEXUS-1 warns at it and clamps the model for stability.

1.3 Delayed neutrons — why a reactor is controllable at all

Over 99% of fission neutrons are *prompt* (born within $\sim 10^{-14}\text{s}$). If they were all there was, a reactor’s response time would be the prompt generation time Λ , which in the NEXUS-1 model is $\Lambda = 2 \times 10^{-5}\text{s}$ — far too fast for any human or rod drive to control. The reactor is controllable only because a small fraction $\beta \approx 0.65\%$ of neutrons are *delayed*: emitted seconds to minutes later by decaying fission products. NEXUS-1 models six delayed groups with decay constants λ from 0.0124 to 3.01s^{-1} . Those slow neutrons set the achievable rate of change of power — which is why staying below \$1 keeps the plant in human-controllable territory.

1.4 Reactor period and startup rate

Reactor period (Reactor Period, seconds) is the time for power to change by a factor of e (≈ 2.72). A long period means slow, gentle change; a short period means power is moving fast.

- Steady power $\Rightarrow$ period = ∞ (NEXUS-1 prints ∞s).
- A positive reactivity step shortens the period (power climbs); a negative step lengthens it or makes it negative (power falls).
- **Startup Rate (SUR)** on the Neutronics screen is the same idea in decades per minute (DPM): how many factors of ten the power changes per minute. Real startups are limited to roughly +1

DPM.

1.5 Reactivity feedback: the reactor pushes back

A reactor is largely self-stabilising because temperature and fission products feed back on reactivity. Three feedbacks dominate, and all appear in the NEXUS-1 reactivity balance:

- **Doppler (fuel-temperature) coefficient.** As fuel heats, resonance absorption in ^{238}U broadens and captures more neutrons — prompt, strong, and always negative. This is the reactor’s fast safety brake. The Kinetics solver uses $-2.5 \text{ pcm}/^{\circ}\text{C}$.
- **Moderator-temperature coefficient (MTC).** As coolant heats it becomes less dense and moderates less effectively; in a properly designed PWR this is also negative. The Kinetics solver uses $-15 \text{ pcm}/^{\circ}\text{C}$ of coolant.
- **Xenon-135.** A fission product and ferocious neutron poison (equilibrium worth $\sim 2700 \text{ pcm}$ here), governed by coupled iodine/xenon decay. It builds and burns off over *hours*, producing the slow power swings and the post-shutdown “xenon dead-time” that operators plan around.

Model honesty – read this

The Neutronics screen displays *different* feedback numbers (MTC $-28/^{\circ}\text{C}$, Doppler $-2.4/^{\circ}\text{C}$) from the Kinetics solver above. Both are textbook-order and representative; neither is fitted to a specific reactor. When you want the numbers that actually drive a simulated transient, read them off the **Reactor Kinetics** screen.

1.6 The PWR plant: two loops, one job

NEXUS-1 models a pressurised-water reactor (PWR). Heat makes its way from fuel to grid through two separated water systems:

1. **Primary loop.** Water at $\sim 155 \text{ bar}$ (kept liquid by the **Pressurizer**) is pumped by reactor coolant pumps (RCP-1A, RCP-1B) through the core, leaving by the *hot leg* (T-hot) and returning by the *cold leg* (T-cold). It never boils and never leaves containment.
2. **Secondary loop.** In the steam generators the primary water boils *separate* secondary water into steam (Main Steam Pressure, Steam Flow), which spins the turbine-generator and is condensed back to feedwater (Feedwater Flow, Condenser Vacuum).

The single most important primary-side number is the **hot-leg/cold-leg temperature difference**: it is proportional to the heat the core is actually producing. Average coolant temperature (T_{avg}) is what the feedbacks above respond to.

1.7 Defence in depth and the safety systems

Safety does not rely on the operator or on software. It relies on independent, hardware, deterministic systems that act without permission:

- **Reactor Protection System (RPS)** — 4 channels, voting logic, trips the reactor (**SCRAM**) on any monitored limit. In the model a SCRAM inserts -8000 pcm of reactivity.
- **ECCS** — emergency core cooling, on **Standby** to flood the core on a loss-of-coolant signal.
- **Containment isolation and emergency diesels (2/2)** — seal the building and guarantee power to safety loads.

The NEXUS-1 authority boundary

Across the whole console, the safety layer is *read-only*. The platform is a monitoring and advisory tool that sits *above* the protection systems and never issues a command *down* into them. Every operator action you can take on these screens is either a simulation input or an

acknowledgement — never a real safety actuation.

1.8 The model's representative constants

These are the literal numbers inside the point-kinetics solver, surfaced on the Model Analysis screen. They are textbook-order and representative — chosen to make the dynamics behave correctly, not to reproduce a named reactor.

Constant	Value	Meaning
Total delayed fraction β	0.006502	sets $\beta \approx 650$ pcm
Prompt generation time Λ	2×10^{-5} s	prompt response timescale
Delayed groups	6	$\lambda = 0.0124 \dots 3.01$ s ⁻¹
Doppler coefficient	-2.5 pcm/°C	per °C fuel temperature
Moderator coefficient	-15 pcm/°C	per °C coolant temperature
Reference fuel temp	600 °C	feedback datum
Reference coolant temp	305 °C	feedback datum
SCRAM worth	-8000 pcm	inserted on trip
Xenon equilibrium worth	~2700 pcm	full-power poison load

2 Part II — Reading the console, screen by screen

The screens below follow the console's own grouping. Full-depth treatment is given to the reactor, heat-transport, safety and diagnostic screens; the supporting modules (personnel, lifecycle, robotics, access, audit, services) are covered more briefly because their nuclear-engineering content is lighter, not because they matter less.

2.1 A. Plant-level situational awareness

Overview

What you see The top-level mimic: Electrical Output MWe, Thermal Power MWt, Reactor Power %, Availability, the Plant Energy Flow Mimic (Reactor → Steam Gen → Turbine → Generator → Grid), a Subsystem Status list and an Output Trend.

The physics behind it This screen makes the energy chain visible end to end. Thermal power (MWt) is what the core produces; electrical output (MWe) is what reaches the grid after the thermodynamic cycle. The ratio MWe/MWt is the plant thermal efficiency (a PWR sits around one-third), so MWe always trails MWt — that gap is physics, not a fault.

Normal vs. off-nominal At steady state every node tracks reactor power and the subsystem list reads Nominal/Armed/Synced. Watch items (Turbine/Generator Watch • vib) and a non-zero Recent Alarms count are your cue to drill down.

Operator response

Use this as the 5-second health check. If MWe and MWt move together and the status column is clean, the plant is coherent. If one node lags the rest, follow the energy flow to the screen for that subsystem (Core, Coolant, Steam, Power) rather than reacting to the summary.

Plant Fleet

What you see Plant Output, Units Online, Installed Capacity, Grid Frequency 50.00 Hz, the 400 kV bus, and a switchable list of large PWR units and small modular reactor (SMR) modules. Selecting a unit drives every detailed Reactor page.

The physics behind it A modular, scalable plant: total output is the sum of online units, each synchronised to a shared bus. Grid frequency is the shared heartbeat — it only holds at 50 Hz if generation matches demand across the whole bus.

Normal vs. off-nominal Online units glow cyan and contribute MWe; offline modules are isolated and contribute nothing. Frequency parked at 50.00 Hz means the bus is balanced.

Operator response

This is the master selector: the unit you pick here is the unit the Core, Rods, Neutronics, Kinetics and 3D screens describe. Before interpreting any reactor number, confirm which unit is selected — an “anomaly” is often just a different unit in a different state.

Plant 3D View & Training Mode

What you see A 3-D schematic of containments, SMR modules, a shared steam header and a spinning turbine, with toggles for **Auto-rotate**, **Labels** and **Steam flow**; and a **Training Mode** marked **DRILL** that runs scored exercises.

The physics behind it The 3-D view is representative geometry, not a survey model — active units glow and drive the steam flow and turbine, standby units are dark. Training Mode is important: its drills run on the *same* point-kinetics model as the Reactor Kinetics screen, so power, period, temperature and xenon behave identically.

Model honesty – read this

Training Mode is for skills demonstration only. It is explicitly *not* a certified operator-training simulator and confers no qualification. The score reflects control of the NEXUS-1 model, nothing more.

2.2 B. The reactor (full depth)

Core

What you see Core Parameters for a PWR • 157 Assemblies: Core Temp, Pressure, Reactor Power, Rod Insertion, Reactivity (pcm), Core Burnup (18.4 GWd/tU), Boron Concentration (642 ppm), a Control Rod Banks table, and a Core Map coloured by assembly outlet temperature.

The physics behind it This is the integrated state of the fission engine. Burnup (gigawatt-days per tonne uranium) measures how far through life the fuel is; as it rises, fuel reactivity falls and is compensated by *reducing* soluble boron — which is why boron concentration (ppm) trends down across a cycle. The Core Map shows the radial power/temperature shape: real cores peak toward the centre, so hotter central assemblies are expected, not alarming. The flatter that map, the better the power distribution.

Normal vs. off-nominal

Quantity	Normal signature	Investigate when
Reactivity (pcm)	near 0 at steady power	persistently non-zero $\Rightarrow$ power changing
Core Map	smooth centre-peaked gradient	one cell far hotter than neighbours (hot channel / tilt)
Boron (ppm)	slow decline over the cycle	a step change with no rod or power move
Rod insertion	small, near the bank’s normal window	deep insertion at full power (axial-offset penalty)

Operator response

Read the Core screen as the “trial balance”: reactivity near zero confirms the core is in equilibrium. A localised hot cell points you to power distribution (rod patterns, possible assembly issue); a boron step with no operator move points to a chemistry or measurement problem. For the actuators behind these numbers, jump to Control Rods.

Control Rods

What you see Rod Drive Control for RCCA • Bank D, an Insertion % slider with Withdraw/Insert/Auto (34%) and SCRAM, a live Reactivity (pcm) readout (green positive / red negative), a Core Cross-Section showing rods descending into the active core, and live Reactor Power, Thermal Power MWt, Electrical Output MWe and Reactor Period.

The physics behind it Control rods (Rod Cluster Control Assemblies) are neutron absorbers. Inserting them removes reactivity (rod worth, in pcm) and the chain weakens; withdrawing adds reactivity. This screen is the live demonstration of the whole causal chain: *reactivity* → *neutron flux* → *thermal power* → *grid output*. At equilibrium the reactivity readout sits near zero because the feedbacks have re-balanced.

Normal vs. off-nominal A rod move produces a brief reactivity spike (green or red), the period shortens or lengthens, power moves, and feedback returns reactivity toward zero at a new power. Power that will *not* settle, or a reactivity readout stuck far from zero, is the off-nominal shape.

Model honesty – read this

In a real plant the rod command path is hardware-isolated and interlocked. Here it is a training-simulator input with no isolation. Treat the slider as a physics teaching tool, not a model of a real control circuit.

Operator response

Move rods in small steps and watch the period, not just the power. A long period means you are changing power gently and within the delayed-neutron regime; a rapidly shortening period is the warning that you are adding too much reactivity too fast. SCRAM is the all-stop: it slams in −8000 pcm. Note on the trend that power cannot drop instantly after a SCRAM — delayed neutrons leave a decay tail. That tail is correct physics.

Neutronics

What you see The nuclear-instrumentation ranges — Source Range (cps), Intermediate Range (A), Power Range (%) — a Startup Rate (DPM), an Axial Flux Profile, and a Reactivity Balance listing Control Rods, Boron −1180, Moderator Temp Coeff −28/°C, Doppler −2.4/°C, Xenon-135 and Net Reactivity, all marked pcm • illustrative.

The physics behind it A reactor is monitored across many decades of power by three overlapping instrument ranges: source range (counts per second, for startup), intermediate range (a current, log scale), and power range (linear %). The Axial Flux Profile shows *where* in the core the neutrons are: as rods insert from the top, the upper core is suppressed and the flux peak shifts downward — the “axial offset” operators manage to avoid skewed burnup and power peaking. The Reactivity Balance is the ledger of every contribution that must sum to the net.

Normal vs. off-nominal At steady power the net reactivity is ≈ 0 , the power range is flat, and the axial profile is gently centre-peaked. A drifting axial offset, or a balance that will not sum to zero, indicates a developing distribution or feedback problem.

Model honesty – read this

This screen is the *illustrative* reactivity display. Its feedback coefficients (MTC -28 , Doppler -2.4) differ from the rigorous Kinetics solver (-15 and -2.5). Use Neutronics to understand the *structure* of a reactivity balance and the instrument ranges; use Reactor Kinetics for numbers that drive real dynamics.

Operator response

Read the three ranges as a continuous story from shutdown to full power; never trust a single range outside its band. During any power change, watch the axial flux profile as closely as the magnitude — *where* the power is can matter more than *how much*.

Reactor Kinetics — the physics heart of NEXUS-1

What you see The Point-Kinetics Engine with simulated time, a prompt critical warning lamp, live Reactor Power %, Net Reactivity (pcm and \$), Reactor Period, Fuel Temp, an auto-scaled power chart, a Reactivity Balance, and Controls: a Control reactivity (rods) slider with $\pm 10/\pm 50$ steps, Hold critical (0), SCRAM, time-scale (1/10/100/600), Pause/Reset and CSV export.

The physics behind it This is the rigorous model: six-group delayed-neutron point kinetics coupled to a two-node thermal model and the iodine/xenon equations, integrated live. Everything from Part I is operating here at once — a rod reactivity step changes the prompt term, the period responds, fuel and coolant temperatures move, Doppler and moderator feedback push back, and over hours xenon drifts.

Normal vs. off-nominal A small $+10$ pcm step: power jumps slightly (the “prompt jump”), then feedback settles it to a modestly higher level on a long period. The dangerous shape is reactivity approaching $\$1$ (≈ 650 pcm): the period collapses toward the prompt timescale and the **prompt critical** lamp lights.

Model honesty – read this

At and beyond $\$1$ the real behaviour is a prompt-critical excursion. NEXUS-1 *clamps* the reactivity update at 0.95β for numerical stability, so the on-screen runaway is bounded. The lamp is telling you the truth (“this is a runaway condition”); the bounded curve is a modelling choice, not the physics.

Operator response

This is the screen to *learn* the reactor on. Try the author’s suggested sequence: a small $+10$ pcm step (watch prompt jump then feedback settle); then a SCRAM (note the delayed-neutron tail — power cannot fall instantly); then raise the time scale to 600 and watch the xenon transient build over simulated hours. The single habit to build: drive by *period and dollars*, not by power alone.

Model Analysis

What you see A Solver validation page: the literal solver constants as Parameters (JSON), a Live trace (CSV) export, and computed vs reference checks — the inhour and prompt-jump validations run against closed-form references.

The physics behind it This is verification, not operation. The inhour equation relates a stable reactor period to reactivity; the prompt-jump approximation predicts the instantaneous power step

after a small reactivity change. Reproducing both confirms the integrator is faithful to the point-kinetics equations.

Operator response

Treat this as the credibility receipt for the Kinetics screen: it demonstrates numerical fidelity to the equations. It deliberately does *not* claim agreement with any real reactor, because the constants are representative. If you doubt a Kinetics result, this is where you confirm the solver, not the plant.

3D Reactor View

What you see Two side-by-side cores for the selected unit: a **measured** core from (simulated) **sensors** (noisy, slightly lagged) and an **expected model** core (clean), each showing **Power**, **Fuel** temperature and **Rods**; **Core Control** (Rod insertion, Withdraw to critical, SCRAM); and a **Twin Divergence** readout.

The physics behind it This is the digital-twin idea made visual. The left core is what your instruments report; the right is what the model predicts. In a healthy plant they agree within sensor noise. **Sustained divergence between measured and modelled is the earliest warning of a developing sensor fault or a real plant change** — before any single alarm trips.

Operator response

Watch the gap, not either core alone. Transient, jittery divergence is ordinary sensor noise. A divergence that is small but *growing and one-directional* is the signal worth chasing — take it to the Digital Twin screen for the residual trend.

2.3 C. Heat transport and power conversion (full depth)

Coolant / Thermal-Hydraulics

What you see Primary Coolant Loops (2-Loop): Loop A/B Flow with pumps RCP-1A/RCP-1B running, Hot Leg A (T-hot), Cold Leg A (T-cold), secondary Feedwater Flow; a Pressurizer with Level, Pressure, Heaters Auto, Spray Valve Closed; Steam Generator Levels (SG-1/SG-2); and Condenser Vacuum.

The physics behind it This is the primary heat-transport system — the loop that carries fission heat out of the core. The hot-leg minus cold-leg difference ($T_{\text{hot}} - T_{\text{cold}}$) is directly proportional to core power. The **pressurizer** is the system's most safety-critical component: by holding primary pressure (heaters boil water to raise it, spray condenses steam to lower it) it keeps the coolant a sub-cooled liquid so it never flashes to steam in the core. Pressurizer level is the plant's primary inventory gauge.

Normal vs. off-nominal

Quantity	Normal	Investigate when
$T_{\text{hot}} - T_{\text{cold}}$	steady, tracks power	widening/narrowing with no power change
Loop flows A vs B Pressurizer pressure	balanced, both pumps running held at setpoint, spray closed	one loop drops (pump/flow fault) falling pressure (inventory/leak) or spray cycling
Pressurizer level	stable	rising (in-surge) or shrinking (out-surge / leak)
SG levels	steady in the normal band	low (feedwater) or high (overflow, carryover)

Operator response

Diagnose the primary side in this order: pressure (is the coolant still sub-cooled?), then inventory (pressurizer level), then flow balance, then the temperature split. A falling pressure *and* falling level together is the classic loss-of-coolant signature and is the single most important pattern to recognise. A pump trip shows as one loop's flow dropping while the other holds.

Steam & Secondary

What you see Main Steam Pressure, Steam Flow (kg/s), Feedwater Flow (kg/s), Condenser Vacuum (kPa); two U-tube SGs with per-SG Pressure, NR Level, Steam Flow, Status; the Main Steam Header with MSIVs Open and Steam Dump/Bypass Closed; the Turbine Train (Governor Valve, HP/LP Turbines, MSR Outlet Temp 221 °C, Turbine Speed); and a Secondary Flow Mimic.

The physics behind it The secondary side turns heat into rotation. In steady state steam flow out of the SGs must equal feedwater flow in — if it does not, SG inventory is changing. The Governor Valve sets how much steam reaches the turbine and therefore the load demand; the MSR (moisture-separator reheater) dries and reheats steam between the high- and low-pressure turbines for efficiency. The MSIVs (main steam isolation valves) are a containment-boundary safety function — normally open, slammed shut to isolate a steam line break.

Normal vs. off-nominal Healthy secondary: steam flow $\approx$ feedwater flow, stable header pressure, MSIVs open, steam dump closed, turbine at synchronous speed. Steam dump opening or a feedwater/steam mismatch is the signature of a load transient or a developing fault.

Operator response

The key conservation check is steam-out vs. feed-in: a persistent mismatch tells you SG level is drifting before the level gauge looks alarming. If steam dump opens, the plant is shedding heat the turbine isn't taking — pair this screen with Power & Grid to see whether the cause is on the steam side or the electrical side.

Power & Grid

What you see Active Power (MW), Reactive Power (MVar), Generator Voltage (kV), Power Factor, an output trend; and Grid Synchronization on the 400 kV bus: Grid Frequency (Hz), Turbine Speed (RPM), Main Breaker Closed, Sync Status In Sync, Phase Angle +2.1°.

The physics behind it The generator is the bridge between the steam plant and the grid. **Active power (MW)** is real energy delivered; **reactive power (MVar)** supports grid voltage; their relationship is the power factor. Synchronisation is strict: to close the main breaker, generator and grid must match in frequency, voltage and phase — the small Phase Angle shows the controlled lead used to pick up load smoothly.

Normal vs. off-nominal Synced operation: frequency at 50 Hz, breaker closed, phase angle small and steady, MW tracking turbine load. A drifting phase angle, frequency excursion, or breaker change is a grid-side event.

Operator response

Read this together with Steam & Secondary. Electrical load and steam demand are two ends of one shaft: a grid disturbance shows up first here (frequency, phase) and propagates back to steam dump and SG levels. Distinguishing a grid event from a plant event is exactly the value of having both screens side by side.

2.4 D. Safety and radiation (full depth)

Radiation / Safety

What you see Area Radiation Monitors (Containment Interior, Aux Building, Fuel Handling, Turbine Hall, Stack Effluent) in $\mu\text{Sv/h}$ and %-of-limit; a Safety System Status block marked Read-only: Reactor Protection (RPS) 4 channels Armed, SCRAM Availability Ready, ECCS Standby, Containment Isolation Armed, Emergency Diesel Gen Auto • 2/2; plus Containment Pressure, Containment Temp and Sump Level.

The physics behind it This is the defence-in-depth dashboard. Area monitors watch for radioactivity escaping its boundary — the Stack Effluent figure (as % of limit) is the regulated release point. Containment pressure, temperature and sump level are the integrity signals of the final barrier: a rise in any of them can indicate a leak inside containment. The safety-system block confirms the independent protection functions are armed and available.

Normal vs. off-nominal

Signal	Normal	Significance if abnormal
Area monitors	background, Normal	rising dose $\Rightarrow$ contamination / barrier breach
Stack effluent	below limit	approaching limit $\Rightarrow$ release pathway
Containment pressure	atmospheric, steady	rising $\Rightarrow$ leak inside containment
Sump level	Normal	rising $\Rightarrow$ unidentified leakage
RPS / ECCS	Armed / Standby	not-armed $\Rightarrow$ degraded protection

Model honesty – read this

This screen is explicitly a *monitoring/advisory layer with no authority over safety actuation*. The protection systems it displays are hardware-isolated, deterministic and human-supervised; NEXUS-1 watches them, it does not command them. “Armed” here is a status report, not something the platform sets.

Operator response

Correlate, don’t react in isolation. A containment dose rise *with* a containment pressure rise *with* a rising sump level is a coherent leak story and far more meaningful than any one of them. Always confirm RPS/ECCS availability is intact while diagnosing — losing a protection function during an event changes everything.

Alarms & Events

What you see An Alarm Event Management list filterable by Critical / Warning / Info, with Time, Severity, Source, Message and Status; the sidebar badge shows the live active count.

The physics behind it Alarms are the plant’s exception channel. The discipline of an alarm system is *prioritisation*: in a real transient, one root fault sets off dozens of downstream alarms (an “alarm flood”), and the operator’s task is to find the first cause, not to chase every annunciator.

Operator response

Work alarms by severity and by time order. The *earliest* critical alarm is usually nearer the root cause than the loudest or most recent one. When a flood appears, this screen hands off naturally to the Root Cause Graph, which is built precisely to collapse a flood into one origin.

2.5 E. Diagnostics, twin and forensics (full depth)

Digital Twin

What you see Sync Lag (ms), Model Fidelity (97.8%), Signals Mirrored (1842), Max Deviation (1.4%); a Simulation vs Live Plant table of Signal / Live Plant / Twin Model / Deviation / Trend (residual) / Status; and Measured vs Model and Residual vs Tolerance charts. Loop A Flow carries a deliberate slow drift to demonstrate the mechanism.

The physics behind it A digital twin runs a physics model in parallel with the live plant and compares them signal by signal. The *residual* is the gap between measured and modelled. The principle, stated plainly on screen: **transient residual is ordinary sensor noise; a sustained, growing residual flags a sensor fault or a real model/plant divergence**. This is condition monitoring before a threshold alarm would ever fire.

Normal vs. off-nominal Healthy: residuals scatter around zero inside tolerance, fidelity high, max deviation small. The watched pattern is a residual that trends one way and *stays* outside tolerance — exactly what the seeded Loop A Flow drift shows.

Model honesty – read this

This is an illustrative twin: the “live plant” is itself simulated, the model is not identified from real plant data, and (per the project’s own design principle) the twin should not be used to measure NEXUS-1’s own internal code divergence. It demonstrates the *method*, not a validated plant model.

Operator response

Click the drifting signal to chart its residual. Ask two questions: is the residual *growing*, and is it *outside tolerance*? Both yes $\Rightarrow$ treat as a real indication and decide sensor-vs-process by cross-checking redundant signals. The twin’s job is to buy you time before a hard alarm.

AI Diagnostics

What you see Two layers. A working Predictive diagnostics demonstrator (failure-probability and anomaly forecasts fed by the live Component Registry); and a DSLM Assistant (Domain-Specific Language Model) marked ROADMAP PLANNED, described as a local, **air-gapped** nuclear-domain model, advisory only, *not running in this build*.

The physics behind it Predictive diagnostics turns component condition into a forward-looking failure probability — prognostics rather than reaction. The planned DSLM is deliberately scoped: served on-site, no external network egress, grounded on plant data, advisory. The architecture choice (air-gapped, interpretable) reflects that anything touching a nuclear plant must be auditable and contained.

Model honesty – read this

Be precise about what is real here: the predictive layer is a Phase-0 demonstrator; the DSLM assistant is documented *target architecture*, not a running model. The screen says so itself — that honesty is the point.

Operator response

Use the predictive layer as a prioritised watch-list: rising failure probability tells you which component to inspect before it fails, complementing the twin’s signal-level view. Treat any future DSLM output as advice to verify against the deterministic screens, never as an authority.

Optimization (RL) — Beta

What you see A Reinforcement Learning page, marked **Beta**, **Twin-only**, **Advisory**. An RL agent learns a rod-control policy by trial and error *inside the digital twin only*, against a fast reduced-order surrogate; the task is to hold target power with minimal rod movement (load-follow with low wear). The learned policy is presented as a *readable table*, not a black box.

The physics behind it Reinforcement learning discovers a control policy by rewarding good outcomes (on-target power, few rod moves) over many simulated episodes. Confining it to the twin means it can explore freely without ever touching a plant; making the policy a lookup table keeps it interpretable — a hard requirement in a safety domain.

Model honesty – read this

Scope discipline matters here. RL is credible only narrowly: as an advisory policy trained in the twin, with no control authority. It does *not* replace the supervised predictive diagnostics, and framing it as “the” control or diagnostic engine would be an overclaim.

Operator response

Read the learned policy table as a *suggestion* for efficient load-follow, then sanity- check it on the Reactor Kinetics screen: does following it keep period long and reactivity well below \$1? If yes, it is a useful efficiency aid; the human and the protection systems remain in charge.

System Dependencies

What you see A System Dependency Graph (click a node to trace downstream propagation with delays), an Influence Matrix (weighted component-to-component coefficients, weak→direct), and a Causal Chain view. Node colours follow the selected unit; dashed violet edges are negative-feedback loops.

The physics behind it Plants are coupled systems: a change in one component propagates, with delay, to others. The graph encodes *who influences whom*; the matrix quantifies *how strongly*. Negative-feedback edges are the stabilising loops (like the temperature feedbacks from Part I) that damp disturbances.

Model honesty – read this

The coefficients scale with unit size and design but are *illustrative, not identified from plant data*. Use the graph for reasoning about propagation structure, not as a calibrated plant model.

Operator response

When something moves, click its node and follow the arrows to predict *what will move next and when*. This is how you get ahead of a developing transient instead of chasing it — and it is the conceptual backbone the Root Cause Graph turns into forensics.

Component Registry

What you see An inventory of plant components with condition/health attributes that feed the predictive diagnostics layer.

The physics behind it This is the asset spine: every monitored component, its health state, and its links into the dependency graph and the AI prognostics. It is the data source that lets “failure probability” mean something.

Operator response

Use the registry as the master list when a diagnostic points at a component: it is where you confirm what the item is, its condition history, and what depends on it. (See the Appendix note on Registry vs. Aging — the Registry is the live asset list; Aging is the degradation-over-time view.)

Trends & History

What you see Historical Trends over 1h/6h/24h/7d for Output, Reactor Power, Core Temp, Pressure, Coolant Flow; scrub, zoom, overlay and export windows for incident review. Backed by a time-series store (PostgreSQL + TimescaleDB).

The physics behind it Many faults are invisible in a snapshot and obvious in a trend: slow drifts, oscillations and creeping offsets only show up against time. Overlaying signals exposes *lead/lag* relationships — which parameter moved first — which is the heart of diagnosis.

Operator response

For any anomaly, widen the window before judging it: a value that looks wrong at 1 h may be a normal daily swing at 24 h. Overlay the suspect signal with its likely cause and read the order of events — the one that moved first is your lead. Export the window straight into Incident Analysis.

Incident Analysis

What you see An Event Reconstruction timeline (e.g. EVT-2026-0418) with timestamped entries from Baseline nominal through Flow oscillation detected, Operator advisory raised, Flow deviation 3σ (Critical alarm) to Recovery; and ranked Root-Cause Hypotheses (e.g. Bearing wear (RCP-1B) 74%, Voltage transient 19%, Sensor fault ruled out 7%).

The physics behind it This is structured post-event forensics: a precise chronology plus a probability-ranked set of causes. The worked example is a reactor-coolant-pump bearing problem — a 0.8 Hz discharge-pressure oscillation that grows to a 3σ flow deviation, is caught by vibration-pattern recognition, trimmed, and recovered, with safety margins explicitly noted as unaffected throughout.

Operator response

Read the timeline first (what happened, in order), then the ranking (why). The 3σ marker is the statistical line where a deviation stops being noise. Note how the advisory was *acknowledged by the operator* — the human stayed in the loop. This screen is where you turn a Trends export into a defensible account of an event.

Root Cause Graph

What you see Three worked incidents — EVT-2026-0418 feedwater/pump cascade, EVT-2026-0419 EMI/sensor artefact, EVT-2026-0420 non-conforming rod / QA escape — on a Causal Fault-Propagation Graph. Edges are typed: solid = engineered fault-tree backbone (PRA/PSA), dashed violet = learned (ML-weighted), dashed orange = artefact (EMI), struck grey = rejected. Active alarms light up nodes; you can click a node to test it as root cause and Replay propagation.

The physics behind it This is the most sophisticated diagnostic on the console. The backbone comes from the plant's fault and event trees (probabilistic safety assessment) and component topology —

every solid edge is auditable. A data-informed layer then re-weights edges, and the three incidents teach three distinct lessons: a genuine mechanical cascade, an artefact that *looks* like a fault but is an EMI/sensor illusion to be discriminated, and a latent manufacturing/QA escape that only surfaces in service.

Model honesty – read this

The graph is *advisory*. The engineered backbone is auditable; the learned weights are an overlay that proposes, the human disposes. Critically, the EMI incident exists to show the system *rejecting* a false fault — a diagnostic that can only ever confirm faults is worse than none.

Operator response

When an alarm flood lights the graph, work backward along solid edges to the origin node, then use the learned weights only to rank competing candidates. Always ask the EMI question: *is this a real process fault or a sensor/EMI artefact?* Test your candidate as root cause and replay the propagation — if the replay reproduces the observed alarm order, you have your cause.

2.6 F. Rod inspection / NDT

Inspection Overview & Rod Radiograph

What you see A QA module: pick a **rod type** and an individual rod, then review its simulated Radiograph (RT) — an X-ray / Gamma film with **Annotations**, **Contrast** and **Zoom** — alongside per-method NDT **Results** and a final **Validation Verdict**.

The physics behind it Every control rod entering the core is checked *non-destructively* — inspected without being damaged. On a radiograph, *brighter* = *denser* material (cladding walls, welds, metallic inclusions) and *darker* = *less* material (voids, porosity, cracks). The goal is to catch a cladding or weld flaw before the rod is ever irradiated, because once in-core a defect cannot be repaired and can release activity.

Normal vs. off-nominal A clean rod shows uniform density and sound welds. Dark spots, linear indications across a weld, or porosity clusters are the flaws the annotations label; they drive the accept/reject verdict.

Operator response

Read the film with the annotations on first to learn the indication types, then off to test yourself. Treat the verdict as the QA gate: a rejected rod never enters the core. This module demonstrates the inspection *reasoning*, not a certified RT technique.

NDT Methods (reference)

What you see A reference table of non-destructive testing methods — Radiography, Ultrasonic (UT), Visual (VT), Eddy Current (ECT), Dye Penetrant (PT), Magnetic Particle (MT) — with what each detects and its typical use, plus a note that rod cladding (zirconium alloy or austenitic stainless) is generally *not* ferromagnetic, so MT often does not apply.

The physics behind it No single method finds every flaw: radiography sees internal volume, UT sizes internal cracks and wall thinning, ECT excels at thin-wall tubing (steam- generator tubes), surface methods (VT/PT/MT) catch surface-breaking defects. Method choice follows the material and the flaw you fear.

Operator response

Use this as the lookup behind the radiograph screen: match the suspected flaw and material to the right method. The MT caveat is the kind of materials-aware detail that signals real domain study — the right tool depends on the metallurgy, not just the geometry.

2.7 G. Supporting operations (brief)

These modules round out a plant operator's world. Their nuclear-physics depth is light, so each gets a short read: what it shows and what to watch.

Personnel — Overview & Stress Test

A **Roster** by sector plus an **Absence Stress Test** that simulates scenarios (**Seasonal flu**, **Night skeleton crew**, **Targeted: lose SS + RO, Pandemic**) and checks whether each sector still meets its **minimum shift complement** with every critical role covered. *Why it matters:* in a real plant a minimum-staffing breach triggers **Technical Specification** action statements — staffing is a licensed safety requirement, not an HR convenience. **Watch:** any sector dropping below complement or a critical role (e.g. Shift Supervisor, Reactor Operator) left uncovered.

Plant Lifecycle — Aging, Decommissioning, Waste

Aging & Degradation tracks how components wear (embrittlement, fatigue, corrosion) over operating life; **Decommissioning** moves a unit cradle-to-grave (a running reactor must be shut down first); **Waste & Spent Fuel** shows a fleet inventory driven by each unit's cumulative burnup, with a decay-heat/activity model. *Key physics:* spent fuel must cool in the pool (here ~5 yr, the amber line) before dry-cask storage, and the long activity tail is dominated by Cs-137. **Honesty note:** spent-fuel and activity figures are representative decay models for demonstration, not a real inventory.

Robotics & Vehicles

A robotic-asset fleet with a **Mission Readiness** test: each mission needs a capable robot that is available, charged, and has *dose budget* remaining — robot electronics have a cumulative radiation tolerance, just like the humans whose exposure is limited. **Watch:** **Dose-exhausted** or **Severe accident** scenarios where missions can no longer be covered — the nuclear-specific constraint here is radiation dose, not just battery charge.

Zone Access — Live Presence & Permissions Matrix

A simulated RFID/GPS access-control model: tagged people, vehicles and robots in plant zones, with a **Permissions Matrix** restricting high-restriction zones (containment, vital area) to specific cleared classes. A **violation** is an entity detected where its clearance does not permit. *Why it matters:* access control to vital areas is a nuclear-security requirement. **Watch:** any **Access Violation** count above zero.

Compliance / Audit

An **Immutable Audit** log: append-only and *hash-chained*, where every operator action and system event is written once and each seal references the previous one, so any alteration breaks the chain. *Why it matters:* nuclear operations demand a tamper-evident regulatory record. **Watch:** a broken seal chain would invalidate the record — the whole point is that it cannot be quietly edited.

Security / Services

The platform’s own health: **Microservice Health** states, and an **OT Security Posture** built on the **Purdue model** — safety at L0/L1 is air-gapped, control L2 is a read-only tap, and data flows *up* the zones through one-way gateways with *no command path back down* to the safety layer. *Why it matters:* this is the architectural guarantee behind every “advisory / read-only” note in this guide. **Watch:** a **Degraded** service (the example shows Incident Analysis re-indexing) tells you which capability is temporarily reduced.

3 Appendix A — Quick-reference: normal operating bands

A consolidated “what good looks like” for the screens an operator scans most. Values are the NEXUS-1 model’s representative figures, not licensing limits.

Parameter	Steady-state signature	First sign of trouble
Net reactivity	≈ 0 pcm	persistent non-zero; approaching \$1 (~ 650 pcm)
Reactor period	∞ (steady)	shortening rapidly toward prompt timescale
Reactor power	flat at demand	moving with no rod/load change
Fuel temp	near 600 °C datum	climbing with power excursion
$T_{\text{hot}} - T_{\text{cold}}$	proportional to power	changing with no power move
Loop A vs B flow	balanced, pumps running	one loop dropping
Pressurizer pressure	at setpoint, spray closed	falling (inventory/leak)
Pressurizer level	stable	shrinking with falling pressure (LOCA shape)
SG levels	steady normal band	low (feedwater) or high (carryover)
Steam vs feedwater flow	equal	mismatch $\Rightarrow$ SG inventory drifting
Grid frequency / phase	50 Hz, small steady angle	excursion $\Rightarrow$ grid event
Area radiation	background, Normal	rising dose $\Rightarrow$ barrier/contamination
Containment pressure	atmospheric	rising $\Rightarrow$ leak inside containment
RPS / ECCS	Armed / Standby	not available $\Rightarrow$ degraded protection
Twin residuals	scatter about zero, in tolerance	growing, one-directional, out of tolerance

4 Appendix B — Rigorous vs. illustrative: which screen is which

The single most important thing to keep straight when judging this console.

Screen	Character	What that means
Reactor Kinetics	Rigorous	live six-group point kinetics + thermal + xenon; the physically meaningful model
Model Analysis	Rigorous	validates the solver against closed-form references
Training Mode Neutronics	Rigorous Illustrative	reuses the Kinetics solver; not a certified trainer reactivity-balance <i>display</i> ; feedback numbers differ from the solver
Digital Twin	Illustrative	demonstrates the method; “live” plant is simulated, model not data-identified
System Dependencies	Illustrative	propagation structure real; coefficients not identified from data
AI Diagnostics (DSLM)	Planned	target architecture, documented but not running in this build
Optimization (RL)	Beta, twin-only	advisory policy trained in the twin; no control authority
Root Cause Graph	Advisory	auditable backbone + learned overlay; proposes, human disposes
Waste / Aging / Access Safety & Radiation	Representative Read-only	decay/wear/tracking models for demonstration monitors hardware protection; never commands it

A note on Component Registry vs. Aging

They are complementary, not duplicates. The **Component Registry** is the live asset list — what exists, its current health, and what feeds the predictive diagnostics. **Aging & Degradation** is the time axis on those same assets — how condition trends over operating life. Registry answers “what is it and how is it now?”; Aging answers “where is it headed?”

5 Appendix C — Where to read more

This guide is the interpretation layer. For the other layers:

- **Functional Guide** — screen-by-screen description of what each page does and where the controls are.
- **Glossary** — ~100 terms across the domain, for non-engineer readers.
- **Reactor Physics Reference** — the equations and constants behind the Kinetics and Neutronics models, including the dual-physics inconsistency and its unification roadmap.
- **Root-Cause Methodology** — the reasoning behind the three worked incidents on the Root Cause Graph.
- **Project Roadmap** — the path from this demonstrator to a production .NET/C#, Angular, SignalR, Kafka microservices architecture.